

Review

Genetic Markers Associated with Ferroptosis in Cardiovascular Diseases

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Abstract

Recently, a number of new genes (*NFE2L2*, *HFE*, *HMOX*, *HIF-1 α* , *ALOX5*, *GPX4*, *PTGS2*, and *IL-6*) have been recognized as playing a role in ferroptosis and genetic predisposition to cardiovascular diseases (CVDs). Identifying these novel genes may facilitate the discovery of therapeutic agents and improve the clinical evaluation of phenotypes and prognoses in CVD patients. In the future, it will be crucial to develop genetic markers that correlate with clinical outcomes for individuals with CVDs. This review highlights recent developments in ferroptosis research while interpreting how genetic factors may contribute to the pathogenesis of CVDs. Understanding this relationship could be invaluable for predicting disease progression in individual patients, informing suitable medical interventions, and facilitating early diagnosis and treatment. Furthermore, we examine the possible uses of these disorders in diagnosis and the various treatment strategies, along with the associated challenges and existing limitations.

Keywords: ferroptosis; CVDs; genetic predisposition



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1. Introduction

Cardiovascular diseases (CVDs) refer to a group of conditions affecting the heart and blood vessels, which are associated with low survival rates [1]. These disorders represent the primary cause of death and illness worldwide. The Global Burden of Disease (GBD) study from 2019 reported that fatalities due to CVDs rose from 12.1 million in 1990 to 18.6 million in 2019, with these deaths constituting 32% of all global mortality, mainly attributable to ischemic heart disease and stroke [2]. CVDs encompass various conditions, including ischemic heart disease, myocardial infarction (MI), unstable angina, stroke, cardiomyopathy, and arrhythmias [2,3].

Key risk factors for CVDs—such as high blood pressure, dyslipidemia, diabetes, air pollution, tobacco use, and obesity—are modifiable [4,5]. This group of disorders arises from a combination of genetic predispositions, lifestyle choices, and environmental influences [2,6,7]. Molecular genetics significantly contributes to the diagnosis, prevention, and management of CVD. Most cardiovascular diseases and their associated risk factors have polygenic origins. Therefore, there is an interaction between environmental and lifestyle risk elements with several polymorphic risk alleles involved [8].

Cardiovascular genomics focuses on the genetic aspects of diagnosing CVDs [9]. This field employs genetic testing to pinpoint specific DNA regions that contribute to or increase the risk of CVD [10].

Over the past 15 years, the significance of common genetic variants in cardiovascular conditions, including single-nucleotide polymorphisms (SNPs) with a minor allele frequency of 5% or higher, has become increasingly apparent due to advancements in genomics [8]. Genome-wide association studies (GWAS) have identified up to 32 genetic loci linked to stroke [11], over 160 loci related to coronary artery disease [12–14], along with additional traits such as intracranial aneurysm [14] (17 loci) and blood pressure [15–18] (>1000 loci). The estimated heritability for coronary artery disease (CAD) and stroke is approximately 40% to 50% [7].

Previous research has demonstrated that various types of regulated cell death (RCD), including apoptosis, pyroptosis, and necrosis, can lead to damage in the heart and blood vessels, thereby interfering with their normal functions and contributing to the onset of cardiovascular diseases (CVDs) [19].

Ferroptosis is a recently identified form of cell death that may significantly impact the progression of CVD. Genetic factors influencing iron metabolism and lipid degradation are implicated in ferroptosis [20]. While ferroptosis has been extensively investigated in cardiovascular diseases, the impact of SNPs on ferroptosis in different cardiovascular diseases remains to be studied. This will enable us to gain a better understanding of the treatment of cardiovascular diseases associated with ferroptotic cell death. Thus, in this review, we provide an overview of recent developments in the study of ferroptosis, with a particular focus on the potential genetic contributions to its role in the pathogenesis of CVDs. Variants in genes can alter either the expression or functionality of proteins associated with ferroptosis, thereby potentially affecting individual susceptibility to and progression of cardiovascular conditions.

The present review aims to discuss genetic evidence linking ferroptosis with the development and susceptibility to CVDs, such as coronary artery disease (e.g., myocardial infarction, stable angina, premature and perimenopausal coronary artery disease), stroke (e.g., ischemic stroke, aneurysmal subarachnoid hemorrhage), hypertension, and cardiomyopathy.

2. Literature Screening Methods

A literature search was performed between November 2024 and March 2025. No time restrictions were imposed as a search criterion. The PubMed, ScienceDirect, Google Scholar, Scopus, MDPI, and Web of Science databases were utilized, using a combination of search terms: genetic variants of CVDs, ferroptosis, and CVDs; SNPs and CVDs. Inclusion criteria were the use of the English language and publications concerning genetic variants of ferroptosis in cardiovascular diseases. Emphasis was placed on identifying genetic variants, SNPs, with validated expression using quantitative real-time polymerase chain reaction (qRT-PCR), GWAS, and biological processes modulated by these genetic variants. Experimental analyses of the involved molecular mechanisms and molecular targets were also conducted.

3. Molecular Mechanisms of Ferroptosis and Cardiovascular Diseases

Ferroptosis is a non-apoptotic form of regulated cell death caused by ferrous iron-driven lipid peroxidation-mediated oxidative stress. Iron metabolism is crucial in determining ferroptosis sensitivity by increasing the labile iron pool (LIP) through uptake and ferritinophagy and reducing the activity of ferritin (a storage iron protein) and ferroportin (an iron exportation protein) [21,22]. Lipid peroxides can be produced through enzymatic

and non-enzymatic pathways. The enzymatic pathway involves lipoxygenases (LOX), which react with polyunsaturated fatty acids (PUFAs) to form lipid hydroperoxides; these enzymes utilize ferrous iron as a cofactor [23]. Non-enzymatic lipid peroxidation is a free radical-driven chain reaction that is divided into three stages: initiation, propagation, and termination. The initiation stage begins with the Fenton reaction, which utilizes ferrous iron and hydrogen peroxide (H_2O_2) to generate hydroxyl radicals ($\bullet\text{OH}$) [24]. These reactive oxygen species (ROS) can attack the allylic carbon of PUFAs, forming carbon-centered radicals. In the next stage, the carbon-centered radicals react with oxygen to generate lipid peroxyl radicals, which then react with other lipids to form lipid hydroperoxides and propagate the chain reaction [24]. In the termination stage, two radicals react with each other, producing malondialdehyde (MDA) and 4-hydroxynonenal (4-HNE) as byproducts, which are highly reactive aldehydes that induce significant damage to biomolecules, promoting cell death called ferroptosis [25]. To prevent lipid peroxidation in ferroptosis, the system Xc-glutathione (GSH)-glutathione peroxidase 4 (GPX4) pathway reduces lipid hydroperoxides to lipid alcohols [26].

Normal cardiac contraction requires vast amounts of energy in the form of adenosine triphosphate (ATP), mainly obtained through mitochondrial oxidative phosphorylation (OXPHOS) [27]. OXPHOS is fed through the electron transport system (ETS) by the tricarboxylic acid cycle (TCA) and fatty acid oxidation (FAO) [28]. In TCA, the reducing agent's nicotinamide adenine dinucleotide (NADH) and flavin adenine dinucleotide (FADH₂) are synthesized to enter the ETS, where they function as electron donors, promoting OXPHOS. Upon entering the ETS, these molecules generate a proton gradient between the inner and outer mitochondrial membranes [29]. This proton gradient is utilized by ATP synthase to produce ATP in the OXPHOS process, where the final electron acceptor is oxygen (O_2). While O_2 functions as the final electron acceptor in OXPHOS, it also serves as an electron acceptor for electrons that leak during this process [30]. Note that when mitochondria have OXPHOS coupled under normal conditions, ETS can release up to 4% of electrons. This leak reduces O_2 to the superoxide radical ($\text{O}_2^{\bullet-}$) as a secondary product of ETS, triggering the production of new ROS, such as H_2O [31]₂. ROS levels act as secondary messengers, activating several signaling redox pathways and cellular enzymes [32]. However, in pathological conditions like CVDs, mitochondria are decoupled, and the leakage of electrons can increase to 24%, promoting oxidative stress and even oxidative damage [33]. The latter is because $\text{O}_2^{\bullet-}$ and H_2O_2 may react together or with different metals, such as ferrous ions (Fe^{2+}), to produce $\bullet\text{OH}$, during Fenton and Haber-Weiss reactions. $\bullet\text{OH}$ is highly reactive and oxidizes DNA. $\text{O}_2^{\bullet-}$ can also react with nitric oxide ($\text{NO}^\bullet$), producing peroxynitrite (ONOO^-). Both ONOO^- and $\bullet\text{OH}$ react with lipids, producing lipid radicals and chain lipid peroxidation [31]. Lipid peroxidation can produce MDA and 4-HNE, which form adducts directly with proteins and DNA, thereby inducing oxidative damage and ferroptosis [34]. To mitigate oxidative damage and ferroptosis, cells utilize antioxidants to neutralize ROS. Antioxidants encompass both enzymatic and non-enzymatic types [35]. For example, enzymes like superoxide dismutase (SOD) and dual oxidase (DUOXs) convert $\text{O}_2^{\bullet-}$ to H_2O_2 , which is further reduced to water (H_2O) by catalase (CAT) and glutathione peroxidase (GPX) [36]. Non-enzymatic antioxidants, such as glutathione (GSH), serve as cofactors for the activity of enzymatic antioxidants, including GPX. Both enzymatic and those related to GSH production are activated by different transcription factors that respond to oxidative stress, including nuclear factor erythroid 2-related factor 2 (Nrf2) and forkhead box O (FoxO) [37]. However, oxidative damage cannot be avoided if ROS levels exceed the antioxidant capabilities, thereby promoting ferroptosis [34]. Regarding ferroptosis in CVDs, Fang et al. [38] showed that ferroptosis contributes to a substantial portion of cell death in cardiomyopathy and ischemia/reperfusion injury (IRI), surpassing even cell death

by apoptosis, necrosis, and autophagy. This group of researchers also demonstrated that ferroptosis inhibition improves the survival of mice with cardiomyopathy treated with doxorubicin [38]. Their research showed that Fer-1, an inhibitor of ferroptosis, reduced doxorubicin-induced mortality. Nevertheless, mortality was not significantly depleted in mice treated with necrostatin-1 (Nec-1, a specific inhibitor of necroptosis), emricasan (an apoptosis inhibitor), or 3-methyladenine (3-MA, an autophagy inhibitor). This group also investigated the role of ferroptosis in IRI, finding that ferroptosis is a significant form of cell death in IRI. Still, it can be prevented using ferroptosis inhibitors such as Fer-1 or dexrazoxane (DXZ, an iron chelator) [38]. The mechanism discovered by this group of researchers was that overexpression of heme oxygenase (HO-1) during IRI induced heme degradation, resulting in iron overload. This iron overload promoted the accumulation of ROS (via the Fenton and Haber-Weiss reactions), leading to lipid oxidation and subsequent ferroptosis [38].

Interestingly, the latest evidence is supported by other research: for example, Gao et al. demonstrated that transferrin transport and glutaminolysis promote ferroptosis triggered by complete amino acid or cysteine deprivation. Due to this group's identification of glutaminolysis as a ferroptosis inducer, they used compound 968, a glutaminolysis inhibitor, to reduce myocardial IRI in an ex vivo cardiac model [39]. Indeed, they found that at the end of reperfusion, hearts treated with compound 968 or deferoxamine (DFO, an iron chelator and ferroptosis inhibitor) improved cardiac function. Ferroptosis is significantly more present in myocardial reperfusion than in ischemia, where markers of ferroptosis in ischemic injury such as acyl-CoA synthase long chain family member 4 (ACSL4), GPX4, iron, and MDA were not significant, but not in myocardial reperfusion, which showed elevated levels of ACSL4, iron, and MDA associated with decreased GPX4 [39]. Interestingly, the reduction of ferroptosis with DFO was observed in rat hearts subjected to I/R compared to ischemia-treated rat hearts [40].

On the other hand, it has been demonstrated that during myocardial I/R, ubiquitin-specific protease 7 (USP7) promotes p53 deubiquitination, activating transferrin receptor 1 (TfR1, a specific marker of ferroptosis), and augmenting cellular iron uptake [41]. However, USP7 inhibition leads to ubiquitination and decreased levels of p53 and TfR1, thereby decreasing iron overload and increasing GPX4, which in turn reduces ferroptosis. In addition to cardiomyopathy and IRI, ferroptosis is present in other CVDs such as atherosclerosis, heart failure, heart valve disease, cardiac arrhythmia, myocarditis, aortic aneurysm, and aortic coarctation [42]. These ferroptosis-related diseases are characterized by iron overload, ROS production, oxidative stress, lipid peroxidation, oxidative damage, and ferroptosis death, which is alleviated by inhibitors of this cell death, such as Fer-1 and DXZ [42].

4. Genetic Variants and Ferroptosis Pathways in CVDs

Cardiovascular genetics is a crucial research tool that enables us to understand disease mechanisms better and develop innovative prevention and treatment strategies. It is essential to note that understanding genetic predisposition enables preventive actions, such as making lifestyle changes or seeking closer medical follow-up, even before symptoms appear.

The mechanism of ferroptosis is involved in the onset and progression of myocardial infarction and other CVDs, as indicated above and by Huang et al. [43]. Recent evidence suggests the identification of differentially expressed ferroptosis-related genes (*NFE2L2*, *ALOX5*, *GPX4*, *HMOX1*, *ATM*, and *HFE*), which play a key role in cardiac tissue function. Genetic variants or SNPs associated with the development and susceptibility to CVD are summarized below. Figure 1; Table 1.

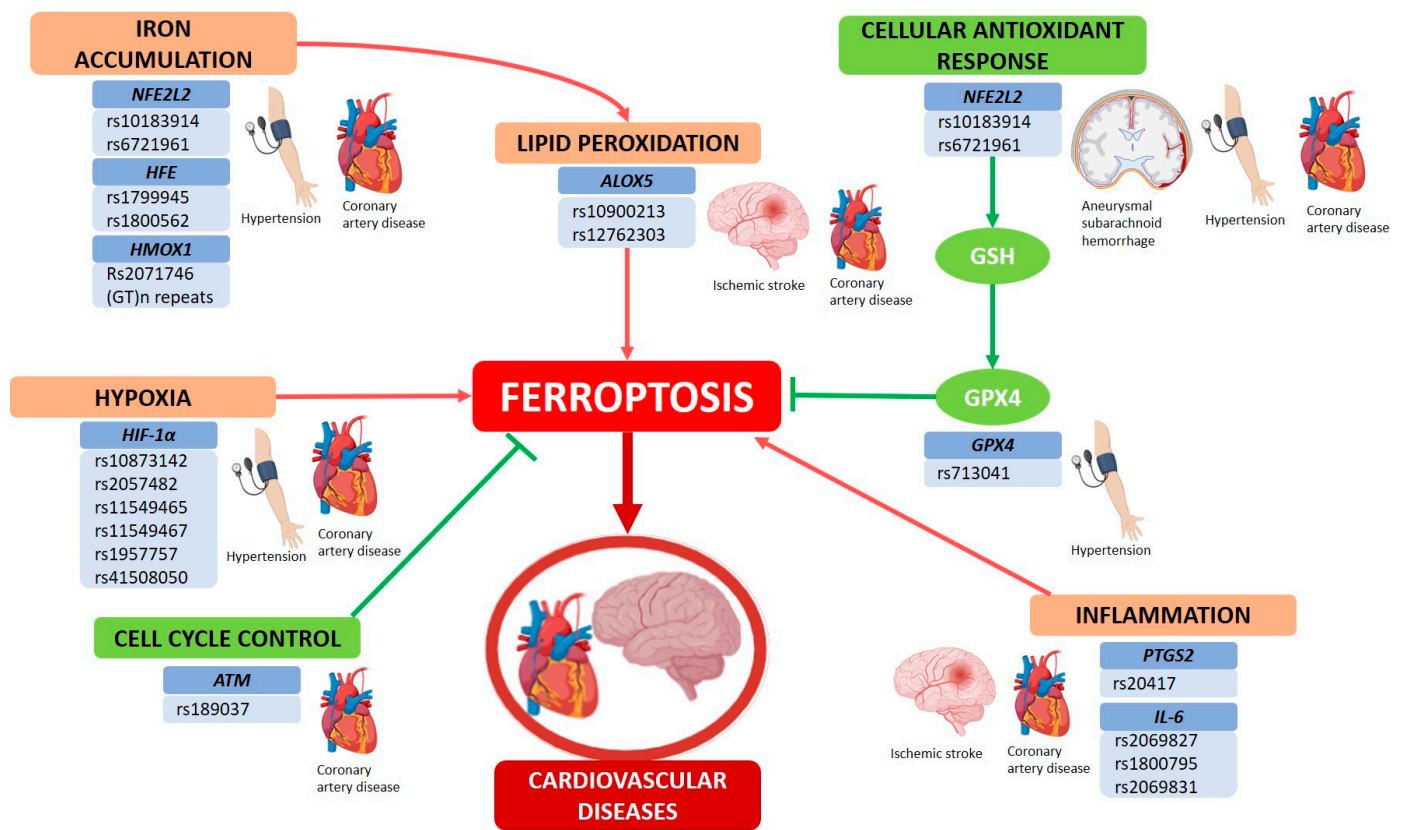


Figure 1. Genetic Variants of Ferroptosis in Cardiovascular Diseases. Ferroptosis-related genes are involved in various signaling pathways that contribute to the pathophysiology of cardiovascular diseases (CVDs), including iron accumulation, lipid peroxidation, hypoxia, inflammation, cellular antioxidant response, and cell cycle control. Single-nucleotide polymorphisms (SNPs) of these genes could increase or decrease the risk of CVDs such as coronary artery disease, ischemic or hemorrhagic stroke, and hypertension. *ALOX5*, arachidonate 5-lipoxygenase; *ATM*, ataxia telangiectasia mutated; *GPX4*, glutathione peroxidase 4; *HFE*, homeostatic iron regulator; *HIF-1α*, hypoxia inducible factor 1 subunit alpha; *HMOX1*, heme oxygenase 1; *IL-6*, interleukine-6; *NFE2L2*, NFE2 like bZIP transcription factor 2; *PTGS2*, prostaglandin-endoperoxide synthase 2.

Table 1. Genetic variants of ferroptosis molecules are associated with CVDs.

Gene	Official Name	Pathway Associated with Ferroptosis	SNP	Cardiovascular Disease	OR	Reference
<i>NFE2L2</i>	NFE2 like bZIP transcription factor 2	Cellular antioxidant response to iron accumulation and lipid peroxidation	rs10183914	Aneurysmal subarachnoid hemorrhage	Risk OR 1.33	[44]
			rs6721961	Blood pressure	Risk ***	[45]
				Coronary artery disease	Risk OR 5.07	[46]

Table 1. Cont.

Gene	Official Name	Pathway Associated with Ferroptosis	SNP	Cardiovascular Disease	OR	Reference
HFE	Homeostatic iron regulator	Iron accumulation	rs1799945	Arterial hypertension	Risk 2.53	[47]
				Coronary heart disease	Risk 1.06	[48]
			rs1800562	Iron status on coronary artery disease	Protective ***	[49]
HMOX1	Heme oxygenase 1		rs2071746	Perimenopausal coronary artery disease	Protective OR 0.67	[50]
			(GT)n repeat (S > L)	Perimenopausal coronary artery disease	Risk OR 1.34	[50]
HIF-1 α	Hypoxia inducible factor 1 subunit alpha		Hypoxia	rs10873142	Perimenopausal coronary artery disease	Protective OR 0.56
		Perimenopausal coronary artery disease **			Risk OR 1.24 **	[50]
		Myocardial infarction			Risk OR 1.28	[51]
		Coronary disease			Risk ***	[52]
		rs2057482		Perimenopausal coronary artery disease	Protective OR 0.71	[50]
				Premature coronary artery disease	Protective OR 0.616	[53]
				Myocardial infarction	Risk OR 1.27	[51]
		rs11549465		Myocardial infarction	Protective OR 0.84	[51]
				Hypertensive left ventricular hypertrophy	Risk OR 1.26	[54]
				Coronary artery disease	Risk ***	[52]
		rs11549467		Hypertensive left ventricular hypertrophy	Risk OR 1.21	[54]
		rs1957757		Hypertensive left ventricular hypertrophy	Risk OR 1.20	[54]
		rs41508050		Coronary disease	Risk ***	[52]
ALOX5	Arachidonate 5-lipoxygenase	Lipid peroxidation	rs10900213	Ischemic stroke	Risk OR 1.991	[55]
			rs12762303	Coronary artery disease	Risk OR 1.36	[56]

Table 1. Cont.

Gene	Official Name	Pathway Associated with Ferroptosis	SNP	Cardiovascular Disease	OR	Reference
<i>GPX4</i>	Glutathione peroxidase 4	Cellular antioxidant response	rs713041	Arterial hypertension	Risk OR 4.19	[57]
<i>PTGS2</i>	Prostaglandin-endoperoxide synthase 2		rs20417	Cerebral infarction	Risk OR 2.84 *	[58]
				Ischemic stroke	Risk OR 1.94	[59]
				Ischemic stroke	Risk OR 1.11	[60]
				Coronary artery disease	Protective OR 0.80	[60]
<i>IL-6</i>	Interleukin -6	Inflammation	rs2069827	Obesity and premature coronary artery disease	Protective OR 0.57	[61]
				Hypertriglyceridemia on premature coronary artery disease	Protective OR 0.72	[61]
			rs1800795	Obesity and premature coronary artery disease	Protective OR 0.40	[61]
				Hypertriglyceridemia on premature coronary artery disease	Protective OR 0.57	[61]
				Cardiovascular disease risk factor	Risk ***	[62]
			rs2069831	Left ventricular ejection fraction	Risk ***	[63]
<i>ATM</i>	Ataxia-Telangiectasia Mutated	Cell cycle control	rs189037	Coronary artery disease	Protective OR 0.47	[64]

* Interaction with other polymorphisms in other genes. ** Interaction with other polymorphisms in the same gene. *** Odds ratio not reported. *IL*, interleukin; *ALOX5*, arachidonate 5-lipoxygenase; *ATM*, ataxia telangiectasia mutated; *GPX4*, glutathione peroxidase 4; *HFE*, homeostatic iron regulator; *HIF-1 α* , hypoxia inducible factor 1 subunit alpha; *HMOX1*, heme oxygenase 1; *IL-6*, interleukine-6; *NFE2L2*, NFE2 like bZIP transcription factor 2; *PTGS2*, prostaglandin-endoperoxide synthase 2; SNP, single-nucleotide polymorphisms, OR, odds ratio.

In the field of genetic epidemiology, GWAS examines numerous prevalent genetic variants across diverse individuals to determine if any of these variants are associated with specific traits. These studies primarily concentrate on the relationships between SNPs and conditions such as significant diseases [65]. Studies with GWAS have revealed many genetic variants that predispose people to different complex diseases. Genes such as *NFE2L2*, *HFE*, *HMOX*, *HIF-1 α* , *ALOX5*, *GPX4*, *PTGS2*, and *IL-6* have been involved in ferroptosis, which contributes to genetic susceptibility to CVDs.

Below, we discuss some of these genetic variants associated with the ferroptosis pathway and the development and susceptibility of cardiovascular disease.

4.1. NFE2L2

NFE2L2 is a gene located on chromosome 2 (2q31) that encodes Nrf2, a transcription factor that upregulates antioxidant genes to protect the cells against oxidative stress. Nrf2 can bind to the promoter regions of *SOD*, *GPX4*, and *peroxiredoxin (PRDX)* genes, thereby preventing lipid peroxidation and, consequently, ferroptosis.

We decided to include the *NFE2L2* gene because it encodes the transcription factor that regulates the expression of genes involved in cellular antioxidant responses, iron metabolism, and lipid metabolism. These cellular processes are crucial for either promoting or inhibiting ferroptosis [66].

For instance, in the aneurysmal subarachnoid hemorrhage (aSAH), a non-ischemic stroke, the brain injury is caused by cerebral vasospasm, inflammation, and oxidative injury. *NFE2L2*, through downstream effectors, can protect the cells from oxidative injury and inflammation after aSAH. In a study by Gaastra B. et al. [44], a SNP, rs10183914 T allele, was associated with a poor outcome following aSAH, with an odds ratio (OR) of 1.33 (Table 1). The cohort included 1069 patients with aSAH from 22 neurosurgical centers in the United Kingdom between 2011 and 2014.

A functional analysis revealed that the T allele in rs10183914 alters the intron–exclusion ratio of *NFE2L2* in brain tissue, resulting in reduced expression [44]. These results suggest that the genetic variant in the *NFE2L2* gene may increase susceptibility to oxidative stress and inflammation in aSAH, thereby contributing to a poor prognosis.

For quality control, candidate SNPs were excluded if the genotyping rate for the cohort was below 90%, the minor allele frequency was less than 0.05, or there was a significant deviation from the Hardy-Weinberg equilibrium ($p < 0.0001$). The study conducted an association analysis using dichotomized clinical outcomes based on the modified Rankin scale score (0–6), which included disability and in-hospital mortality, utilizing both a discovery cohort and a validation cohort. The authors suggest that their findings identify *NFE2L2* as a potential therapeutic target in the context of aSAH and various forms of intracranial hemorrhage, thereby supporting ongoing efforts to carry out a GWAS on outcomes following aSAH. A limitation of this research is the absence of analysis regarding mRNA expression levels or protein concentrations. Therefore, further studies are needed to validate the role of rs10183914 in regulating *NFE2L2* expression. On the other hand, high blood pressure (BP) doubles the risk of CVDs for every 20 mmHg increase in systolic BP and every 10 mmHg increase in diastolic BP. Studies have shown that oxidative stress in the vessels can increase BP through the degradation of nitric oxide (NO), a vasodilator [24,46]. *NFE2L2* may offer protection from vascular oxidative stress. Shimoyama et al. [45] studied two SNPs located within the antioxidant-responsive element (ARE)-like promoter binding sites region of the *NFE2L2* gene, specifically the rs356652124 G allele and the rs6721961 A allele polymorphisms, in the Japanese population. Both were associated with reduced transcription activity of *NFE2L2* [45]. This study reported that males with the rs6721961 (CA+AA) genotype showed lower systolic and diastolic blood pressure (BP), while female AA carriers exhibited higher diastolic BP. Interestingly, the diastolic BP was significantly low in female subjects with rs356652124 (AG+GG) [45]. Furthermore, the rs6721961 (CA+AA) polymorphism was associated with high levels of iron; this may be due to the *NFE2L2*-mediated expression of heme oxygenase-1 (HMOX1), an enzyme that produces iron as a byproduct of heme degradation [67]. Thus, rs6721961 polymorphism could promote ferroptosis by increasing the *NFE2L2* transcription activity. Nonetheless, the GWAS is still limited to specific populations, and the identification of low-*NFE2L2*-activity polymorphisms may help recognize oxidative stress-sensitive populations that could benefit from antioxidant or anti-ferroptotic therapies. However, further studies are required to determine whether rs6721961 dysregulates blood pressure by decreasing *NFE2L2* levels and their downstream targets.

Coronary artery disease (CAD) is a chronic, multifactorial disease and the major contributor to CVD mortality [4]. The principal cause of CAD is atherosclerosis, a process that involves the accumulation of oxidized low-density lipoproteins (ox-LDL), inflammation, endothelial dysfunction, and oxidative stress. A study by Sarutipai boon et al. [46]. included

393 Thai subjects divided into two groups based on coronary angiography findings: a CAD group (with >50% coronary artery stenosis) and a non-CAD group (with <50% coronary artery stenosis). The study also includes mRNA expression of *NFE2L2*. They observed that the TT genotype was associated with lower *NFE2L2* mRNA expression levels and greater atherosclerosis severity (as measured by the Gensini score) in CAD patients. This provides a mechanistic insight into SNPs in the *NFE2L2* gene and coronary artery diseases; thereby, the rs6721961 polymorphism could be useful as a prognostic marker. However, the Nrf2-regulated genes involved in atherosclerotic plaque formation are still unknown.

4.2. *HFE* (Homeostatic Iron Regulator)

The *HFE* (homeostatic iron regulator) gene is located on chromosome 6 (6p22.2). It encodes the hemochromatosis protein, also known as the homeostatic iron regulator, a membrane protein and MHC (Major Histocompatibility Complex) class I antigen that plays an essential role in iron metabolism as a sensor and regulator of iron levels [67]. The mutations in the *HFE* gene are associated with familial hemochromatosis, a genetic disorder characterized by tissue iron overload, which increases the susceptibility of LDL cholesterol to lipid peroxidation within the arteries, thereby enhancing the risk of atherosclerosis and CAD [49]. In a case-control study [47] involving 1405 subjects, there were 939 patients with arterial hypertension and 466 healthy controls. The population consisted of Caucasians from Central Russia, recruited from 2013 to 2016 at the Cardiology Department of St. Joasaph Belgorod Regional Clinical Hospital. The GG genotype in rs1799945 of the *HFE* gene was associated with an increased risk of arterial hypertension, with an OR of 2.53 (Table 1) [47]. In addition, the rs1799945 polymorphism showed a strong association with iron status, hypercholesterolemia, and high levels of triglycerides; these could promote ferroptosis in patients with arterial hypertension, albeit the mechanism is still unknown, and further studies are required. The limitation of this study was the lack of characterization of gene and protein expression, as well as additional mechanistic studies that demonstrate intergenic interactions with risk alleles in the promoter region of the *HFE* gene. The study's strength lies in the in silico data, which shows the potential function of rs1799945 *HFE* in conjunction with other genes involved in arterial hypertension pathophysiology. Therefore, this study highlights the rs1799945 polymorphism as an SNP that warrants further investigation as a potential therapeutic target for atherosclerosis and CAD [48].

On the other hand, a work by Gill et al. [49] suggests that a higher iron status lowers the risk of CAD by reducing LDL levels; this iron status is favored by the rs1800562 polymorphism in the *HFE* gene. Alternatively, Gill et al. showed that the rs1799945 polymorphism is related to higher systolic and diastolic BP; likewise, in the previous study, these results provide a hint of the relation between rs1799945, iron metabolism, and blood pressure. In addition, a meta-analysis concluded that the *HFE* rs1799945 G allele polymorphism increases the risk of CAD by 6% in the Caucasian population, with an OR of 1.06 (Table 1). On the other hand, the results in the Chinese population showed no significant association between rs1799945 and CAD [48]. Based on the previous data, certain polymorphisms can influence CVD outcomes in specific populations. However, the studies are limited to certain populations. Therefore, it would be interesting to study the genetic variants and their relationships in other populations, as well as in relation to different cardiovascular diseases. Hence, further studies are needed to gain insight into the genetic variants associated with coronary artery disease in specific ethnic groups.

4.3. *HMOX1* (Heme Oxygenase 1)

The *HMOX1* gene encodes heme oxygenase-1, an enzyme that degrades heme, producing biliverdin, carbon monoxide, and iron as byproducts. *HMOX1* plays a dual role in

ferroptosis; under physiological conditions, HMOX1 protects cells against oxidative injury by ROS by upregulating the levels of biliverdin and bilirubin [21]. However, under pathological conditions (iron overload, excessive oxidative stress), HMOX1 increases ferrous iron levels from heme degradation and induces mitochondrial dysfunction, promoting ferroptosis [68,69]. Genetic variations in the *HMOX1* gene could be associated with an increased risk of CVDs. A case-control study in perimenopausal Chinese patients [50] involved 589 CAD patients and 860 controls from the female Chinese population, who were divided into groups based on coronary arteriography results (CAG) and pre- and postmenopausal status. The study revealed that *HMOX1* rs2071746 was associated with a reduced risk of perimenopausal CAD (OR = 0.67) (Table 1). This may be a result of an enhanced antioxidant response within the atherosclerotic plaque promoted by HMOX1.

Furthermore, the rs2071746 AT genotype and TT genotype in premenopausal and postmenopausal subjects, respectively, had lower aspirin resistance in CAD (OR = 3.02; OR = 3.05) (Table 1), which means that those who carried these genotypes and received antiplatelet therapy had a better response than those with the AA genotype. This is particularly important in the pharmacogenomic field, as it enables the improvement of individual treatments, drug development, and prognosis. Interestingly, rs2071746 is located between an extra exon (Exon1a) and Exon1 in the 5'-untranslated region (5'-UTR) of *HMOX1*, where it is involved in alternative splicing of the gene. Additionally, it has been described that the A allele enhances *HOMX1* activity, especially in endothelial cells, and promotes a cardioprotective effect [70]. Nonetheless, the study did not prove whether rs2071746 was associated with altered mRNA expression or protein levels of *HMOX1* in these patients. Therefore, further studies are required to address this question.

Another genetic variant found in the promoter region of the *HMOX1* gene is the (GT)*n* dinucleotide repeat length polymorphism, which is divided into three classes: class S (<25 repeats), class M (26–31 repeats), and class L (≥ 32 repeats). The length of the (GT)*n* repeats regulates the expression of *HMOX1* under oxidative stress, and larger (GT)*n* repeats fail to bind to specific transcription factors, suggesting that L genotype patients could be associated with greater oxidative stress, atherosclerosis, and cardiovascular disease [71]. According to Guo et al. [50], the L allele was associated with a significantly increased risk of CAD in postmenopausal patients (OR = 1.34). This may be caused by reduced *HMOX1* expression, leading to increased oxidative stress that promotes inflammation and the rupture of atherosclerotic plaques in the coronary arteries. Interestingly, (GT)*n* repeat polymorphism was correlated with the amount of vessel lesions. Therefore, studying SNPs in the *HMOX1* gene could lead to improved individualized treatment for cardiovascular diseases.

4.4. *HIF-1 α* (Hypoxia Inducible Factor 1 Subunit Alpha)

The *HIF-1 α* gene is located on chromosome 14 (14q23.2) and encodes the alpha subunit of HIF-1 α . This oxygen-sensitive transcription factor regulates genes involved in angiogenesis, energy metabolism, cell proliferation, glucose metabolism, and iron metabolism, thereby maintaining oxygen homeostasis. *HIF-1 α* can inhibit ferroptosis through various mechanisms, including enhancing cysteine uptake, iron storage, preventing mitochondrial damage, and reducing iron uptake [72]. On the other hand, under certain circumstances, HIF-1 α can increase ferroptosis susceptibility by upregulating HMOX1, which leads to excessive iron and lipid peroxidation [73]. SNPs in the *HIF-1 α* gene could predict susceptibility to coronary artery diseases. The rs10873142 polymorphism is located in the intron region of the *HIF-1 α* gene, where the T allele has been associated with increased transcriptional activity and angiogenesis, suggesting that higher levels of HIF-1 α could reduce the risk of cardiovascular disease [74]. Guo et al. [50] reported a lower risk of CAD in Chinese premenopausal women with the rs10873142 polymorphism compared with post-

menopausal women, with an OR of 0.56 (Table 1). Nonetheless, Li Y. et al. demonstrated in a meta-analysis that the rs10873142 T > C variant was associated with an increased risk of CAD in the Chinese Han population (OR = 1.28) (Table 1) [51]. Since these studies did not measure HIF-1 α serum levels, it would be interesting to determine if high HIF-1 α levels resulting from the rs10873142 polymorphism are involved in CVD outcomes. Another HIF-1 α SNP, rs2057482, is located in the 3'-untranslated region (3'-UTR) of the *HIF-1 α* gene, close to the microRNA binding site. Moreover, an in silico analysis reported that the rs2057482 T allele could regulate the expression of *HIF-1 α* by microRNA 199a [75]. Guo et al. suggested that rs2057482 could regulate susceptibility to CAD in an allelic model (OR = 0.71) [50] (Table 1). However, the expression of *HIF-1 α* was not accounted for in the analysis of whether rs2057482 reduced or increased it. Consequently, additional research is required to strengthen the existing evidence.

Specifically, the rs10873142 and rs2057482 polymorphisms, when combined, have a strong association with the risk of perimenopausal CAD, with odds ratios of 1.24 and 0.71, respectively [50]. Due to these results, the question remains whether a single polymorphism or the combination of multiple polymorphisms in a gene is associated with CVD outcomes. A study in the Mexican population by López-Reyes et al. [53] found that the rs2057482 T allele was associated with decreased cardiovascular risk factors, including obesity, hypertension, hypercholesterolemia, and hypertriglyceridemia, which, in turn, lowers the risk of premature CAD (OR 0.61) (Table 1). However, there was an increased risk for type 2 diabetes (OR = 4.76), a significant cardiovascular risk factor. The role of the rs2057482 polymorphism in glucose metabolism remains to be elucidated. Interestingly, Li Y. et al. [51] found that the *HIF-1 α* rs2057482 was related to high-density lipoprotein (HDL) cholesterol levels in Chinese MI patients but not overall MI or CAD risk. Because of the results mentioned above, Li Y et al. performed a meta-analysis that showed that for the Caucasian population, HIF-1 α rs2057482 C > T polymorphism increased the CAD risk (OR = 1.27) (Table 1); concurrently, the same SNP in the Asian population was associated with reduced CAD (OR = 0.71) (Table 1). It has been suggested that the *HIF-1 α* rs11549465 T allele may alter protein stability and disrupt HIF-1 α function due to an amino acid substitution from proline 582 to serine, which could affect hypoxia response and antiapoptotic pathways [76]. However, Li et al. observed that the rs11549465 T allele was correlated with a reduced CAD risk (OR = 0.84) (Table 1) [51]. Further research in other populations is required, even though the data are primarily restricted to Asians and Caucasians. The results of this study provide additional evidence that a single allele can yield distinct outcomes in different populations.

Sheng et al. [54] studied three SNPs of the *HIF-1 α* gene in the Chinese population with primary hypertensive left ventricular hypertrophy (LVH), a hypertensive organ damage, and an independent risk factor of CVD. They found that the rs11549565 T allele, the rs11549467 G allele, and the rs1957757 T allele were associated with an increased risk of LVH (OR = 1.26; OR = 1.21; OR = 1.20 (Table 1), respectively). The rs11549467 also creates an amino acid substitution from alanine 588 to threonine, which could alter protein–protein interaction under cellular stress [77]. On the other hand, rs1957757 is an intronic SNP, which suggests a transcriptionally regulatory function in the *HIF-1 α* gene [78]. Additionally, the GG genotype of rs11549467 was correlated with an elevated risk of LVH (OR = 1.75) and higher plasma *HIF-1 α* concentrations compared to patients with the GA+AA genotype and controls.

Hlatky et al. [52] genotyped *HIF-1 α* polymorphism in patients with either acute MI or stable exertional angina as their clinical presentation of CAD. They reported that the minor alleles of rs11549465, rs10873142, and rs41508050 were more prevalent in patients with stable exertional angina than those who presented with acute MI [52]. Moreover, the same

SNPs were independent predictors of stable angina rather than acute MI as the initial presentation of CAD. Remarkably, the rs11549465 and rs10873142 polymorphisms are located in regions that regulate the degradation and stability of the *HIF-1 α* mRNA transcript [52]. We hypothesized that a decrease in *HIF-1 α* activity reduces plaque neovascularization and the risk of intraplaque hemorrhage and, thereby, acute MI. To date, the genetic variants in the *HIF-1 α* gene have been the most extensively studied in CVD, and they may serve as effective prognostic biomarkers in specific populations.

4.5. *ALOX5 (Arachidonate 5-Lipoxygenase)*

Arachidonate 5-lipoxygenase (ALOX5) is a non-heme iron-containing enzyme encoded by the *ALOX5* gene [79]; it catalyzes the peroxidation of PUFAs (enzymatic lipid peroxidation pathway), which triggers ferroptosis [80,81]. Genetic studies have shown the correlation between 5-lipoxygenases and atherosclerosis and CVD due to their involvement in inflammation and oxidative stress. The rs10900213 polymorphism is located in the intron 3 region of the *ALOX5* gene, and, therefore, it may regulate protein expression. Liu D. et al. [55] conducted a case-control study that enrolled 351 ischemic stroke patients and 417 controls in the Han Chinese population. The study revealed an epistatic interaction between *ALOX5* rs10900213, *ALOX5AP* (Arachidonate 5-Lipoxygenase Activating Protein) rs4293222, and *MPO* (Myeloperoxidase) rs2107545, which significantly increased the risk of ischemic stroke in the Han Chinese population (OR = 1.99) (Table 1). Nevertheless, there was no significant association between rs10900213 itself and ischemic stroke. One limitation of the study is that the results are limited to cellular levels, not systemic levels. Therefore, other pathways involving *ALOX5* should be explored to evaluate this marker as a prognostic factor in the serum of patients with ischemic stroke.

Interestingly, another study conducted by Heidari et al. [56] involved a cohort of 50 patients with CAD and 50 controls from Iran as part of a case-control study. Another *ALOX5* SNP, rs12762303, was found to be associated with the risk of CAD. The *ALOX5* rs12762303 C allele, an SNP located in the promoter region, was correlated with the risk of CAD. Additionally, the CC + CT genotypes of rs12762303 were associated with higher *ALOX5* mRNA expression levels [56]. The enhanced *ALOX5* expression could increase ferroptosis sensitivity, thereby augmenting inflammatory processes such as atherosclerosis. Nonetheless, this study includes a total population of 100 individuals, while other studies recruited at least 400. Therefore, we encourage larger studies in this field. Although rs12762303-mediated increased *ALOX5* activity may promote inflammation that leads to the occurrence of CAD, rs12762303 could still be used as a prognostic genetic marker.

4.6. *GPX4 (Glutathione Peroxidase 4)*

The *GPX4* gene, located on chromosome 19 (19p13.3), encodes the selenium-containing protein enzyme GPX4, a critical regulator of ferroptosis due to its ability to catalyze the reduction of lipid hydroperoxides into lipid alcohols in the cell membrane, thereby protecting the cell against oxidative damage [27,82]. The genetic variants of the *GPX4* gene could affect gene, protein expression, and function, resulting in oxidative stress, lipid peroxidation, and, thereby, ferroptosis [57]. The 3'-UTR in the *GPX4* gene contains a sequence necessary for the incorporation of selenium and GPX4 protein synthesis; the rs713041 polymorphism is located in this region and modulates its expression [57]. A meta-analysis conducted by Barbosa et al. [57] included four studies on hypertension-related diseases, with a total of 2149 cases and 2250 controls. Two of the studies were performed on the Han Chinese population, and the other two were performed on the European population. The study found that carriers of the *GPX4* rs173041 T allele were associated with elevated risk of developing stroke and arterial hypertension in the European population. Furthermore, the

TT genotype was associated with increased lipid peroxidation and expression of adhesion molecules in endothelial cells, leading to inflammation and elevating the risk of CVDs [57]. Due to the established role of *GPX4* as a critical regulator of ferroptosis, the rs713041 variant could serve as a promising prognostic biomarker in cardiovascular diseases associated with ferroptosis. The presence of the T allele could lead to the development of antioxidant therapies that benefit patients with hypertension-related diseases.

4.7. *PTGS2* (Prostaglandin Endoperoxide Synthase 2)

The *PTGS2* gene, located on chromosome 1 (1q31.1), encodes prostaglandin-endoperoxide synthase 2 (*PTGS2*), also known as cyclooxygenase 2 (*COX-2*), an inducible enzyme that plays a crucial role in inflammation through the biosynthesis of prostaglandins [83]. *PTGS2*, as an inflammatory marker, is increased in atherosclerosis and hence could contribute to plaque rupture and CVDs [3,59,84]. In ferroptosis, the *PTGS2* is significantly upregulated by lipid peroxidation [21]. Although *PTGS2* is not a major driver of ferroptosis, it could be useful as a biomarker of this process [22,85]. The *PTGS2* rs20417 is a functional promoter polymorphism and the most studied polymorphism in cardiovascular diseases, specifically coronary artery disease and stroke. Due to its location, it could affect *PTGS2* expression and, therefore, inflammatory processes [58–60]. Yi X et al. [58] conducted a case-control study, including 411 cases with cerebral infarction and 411 healthy controls in the Han Chinese population. The study showed that Chinese individuals who carried both *ALOX5AP* SG13S114 AA and *PTGS2* rs20417 CC genotypes had an increased risk of cerebral infarction (OR = 2.84) (Table 1), suggesting that the interaction between both enzymes raised the susceptibility to cerebral infarction. Yet, research assessing the expression of proteins in the leukotriene pathway to determine whether these polymorphisms may influence it is still lacking. Understanding the role of the leukotriene pathway in cardiovascular disease could inform the development of therapies focused on inflammation, potentially reducing the incidence of adverse outcomes.

Additionally, another study indicated that the frequency of the rs20417 CC genotype was notably greater in patients presenting with vulnerable carotid plaques and elevated platelet aggregation upon admission (OR = 1.94, $p = 0.035$) (Table 1), which contributes to the pathophysiology of ischemic stroke [59]. Conversely, a meta-analysis revealed that the rs20417 C allele is protective against CAD and not associated with ischemic stroke (OR = 0.80, $p = 0.03$; OR = 1.11, $p = 0.38$, respectively) (Table 1) [60]. Additionally, the CC genotype repressed *PTGS2* promoter activity and conferred a protective effect from CAD; this could be owing to lower *COX-2* levels and, hence, decreased levels of inflammatory mediators, which are involved in atherosclerosis and CAD [60]. It appears that the rs20417 polymorphism confers protection against CAD while also increasing the risk of stroke, indicating a dual role in CVD. However, further studies are needed to determine which type of carriers are susceptible to stroke. The latter is consistent with the results of other studies. However, the study did not account for ethnic differences in studies that could have influenced the results.

4.8. *IL-6* (Interleukin-6)

Interleukin-6 (*IL-6*) is a pro-inflammatory cytokine encoded by the *IL-6* gene, located on chromosome 7 (7p15.3) [86]. Besides its role in inflammation, this cytokine can increase ferroptosis sensitivity by disrupting iron homeostasis through the elevation of cellular LIP and regulation of ferritin synthesis, leading to non-enzymatic lipid peroxidation and oxidative stress [87]. A study in the Mexican population by Posadas-Sánchez et al. observed that rs2069827 and rs1800795 polymorphisms in the *IL-6* gene were associated with specific cardiovascular risk factors in patients with premature CAD (pCAD) and the control group,

respectively. The former with the T allele was associated with a lower risk of central obesity and hypertriglyceridemia in the control group (OR = 0.577, $p = 0.03$; OR = 0.728, $p = 0.03$, respectively) (Table 1) [61]. On the other hand, the rs1800795 C allele was also correlated with a low risk of central obesity and hypertriglyceridemia in the control group (OR = 0.40, $p = 0.01$; OR = 0.57, $p = 0.03$, respectively) (Table 1) [61]. Rs1800795 was associated with increased levels of C-reactive protein (CRP) (OR = 1.49, $p = 0.007$) (Table 1), a biomarker of systemic inflammation, in subjects with pCAD [61]. It is a systemic inflammation biomarker, and the latest literature supports the notion that this type of inflammation is a crucial contributor to acute coronary syndromes (ACS). Therefore, increased CRP serum levels could also serve as a predictor of poor prognosis in ACS [88]. Whether the high levels of CRP are owing to *IL-6* rs1800795 remains elusive, as other cytokines can induce CRP production.

Nonetheless, the mechanism by which the rs1800795 C allele can increase CRP levels and reduce the risk of obesity remains unknown and requires further study. Furthermore, the rs2069827 TT genotype was associated with the null expression of *IL-6* in subcutaneous and visceral adipose tissue, as well as in coronary arteries. Individuals with the rs1800795 C allele had higher *IL-6* serum levels and increased expression in subcutaneous and visceral adipose tissue, as well as in coronary arteries, atria, and the left ventricle [61]. However, they did not observe a correlation between *IL-6* genotypes and pCAD. Interestingly, Tabrez S. et al. [62] observed in the Saudi population that the rs1800795 G allele was associated with higher concentrations of neopterin, another inflammatory biomarker, in patients with CVD. Finally, another polymorphism, rs2069831 with the TT genotype, was correlated with left ventricular ejection fraction (LVEF) and its reduction in patients with hypertrophic cardiomyopathy. Nevertheless, this polymorphism was not associated with serum *IL-6* levels [63].

4.9. Ataxia-Telangiectasia Mutated (*ATM*)

The *Ataxia Telangiectasia Mutated (ATM)* gene encodes a serine/threonine protein kinase that mediates the DNA double-strand break response (DDR) to regulate DNA repair and apoptosis. Only Ding et al. [64] conducted a case-control study that enrolled 190 patients with essential hypertension and 179 control individuals in the Chinese population. The objective of the study was to observe the association between rs189037 and essential hypertension. However, there was no association even after adjustment for confounding factors. On the other hand, they observed that the TT genotype could be a protective factor in CAD patients. The latter is aligned with the results of an in silico study made by Huang et al. [43], who found that *ATM* expression is a protective factor in patients with myocardial infarction. This is the only study examining *ATM* SNPs and CVDs. Therefore, further studies are needed to investigate the role of genetic variants in the *ATM* gene in other cardiovascular diseases, including myocardial infarction, stroke, arrhythmias, and others.

5. Discussion

Pharmacogenomics in cardiovascular disease (CVD) focuses on tailoring treatment to a person's genetic profile to predict their response to medications and enhance therapeutic outcomes. This strategy can assist in pinpointing patients who are more susceptible to negative drug responses or who might not react favorably to specific treatments, ultimately resulting in more effective and safer healthcare solutions.

The association between cardiovascular disease (CVD) and genetic polymorphisms as key regulators of ferroptosis is established but not yet fully understood. This review aims to provide an overview of the current state of knowledge and stimulate further exploration

of the interesting association between genetic variants related to the ferroptosis pathway and cardiovascular disease.

Studying genetic variants is crucial for understanding human health and disease, but it is not without limitations. These include challenges in interpreting the clinical significance of variants, difficulties in detecting certain types of variants, and the potential for false positives or negatives. Additionally, genetic testing may not always predict disease onset, severity, or progression, and it can raise ethical concerns [89].

Additionally, ethical considerations and the complexity of genetic interactions also pose a significant hurdle. Understanding the role of genetic variants in ferroptosis related to CVDs can help identify individuals at higher risk and potentially guide personalized treatment strategies. Targeting ferroptosis pathways with drugs that modulate iron metabolism, reduce oxidative stress, or inhibit lipid peroxidation may offer therapeutic benefits in CVDs.

Further research is needed to fully elucidate the complex interplay between genetic variations, ferroptosis, and CVDs, which could lead to more effective prevention and treatment approaches. The evolution of molecular biology techniques—such as transcriptomics, genomics, and epigenetics—has advanced the field of genetic marker identification. Additionally, progress in bioinformatics and interdisciplinary collaborations has significantly improved our capacity to gather, describe, and analyze vast datasets produced by technological innovations.

Despite the significant discoveries about the role of genetic variants linked to ferroptosis in cardiovascular diseases (CVDs), it is crucial to acknowledge the limitations present in several reviewed studies that may affect the interpretation of their findings. Some of these studies are observational in nature, which limits the ability to determine causal links. Furthermore, many studies utilize *in silico* data, which may lead to biases related to the lack of experimental information. Additionally, variability is noted in the criteria employed to evaluate associations with clinical outcomes.

Ongoing research is needed to identify new genetic markers, validate existing ones, and develop clinical guidelines for personalized cardiovascular treatment.

6. Conclusions

This review represents the initial examination of ferroptosis-related genes associated with cardiovascular diseases (CVDs), which could serve as valuable biomarkers for ferroptosis. These findings also suggest that the ferroptosis-related genes associated with CVDs have significant diagnostic and prognostic significance for several CVDs. However, the evidence shows that certain SNPs are related to CVDs in some ethnic populations. Therefore, more extensive investigations are needed to dissect further the exact implications of ferroptosis in the pathogenesis of cardiovascular diseases. The techniques available for the development of markers are genomics (SNP genotyping, pharmacogenetics, and gene expression analyses), which could provide biomarkers for the diagnosis of several CVDs, along with predicting disease progression and response to therapy. This could be greatly beneficial in predicting the course of disease in individual patients, guiding appropriate medical treatment, and stratifying the patient risk profile.

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